

Chat Link: <https://share.gemini.google/6BoSWFqls2RA>

Updated: <https://share.gemini.google/jgjfHRt4r1YO>

```
global_data = {  
  "attribute_1": "total_cost",  
  "attribute_2": "material_continuity",  
  "attribute_3": "electrothermal_performance",  
  "attribute_4": "mass",  
  "attribute_5": "mechanical_integrity",  
  "direction_attribute_1": "positive",  
  "direction_attribute_2": "positive",  
  "direction_attribute_3": "positive",  
  "direction_attribute_4": "positive",  
  "direction_attribute_5": "positive",  
  "attribute_1_min": 0.20,  
  "attribute_1_max": 0.99,  
  "attribute_2_min": 0.05,  
  "attribute_2_max": 0.98,  
  "attribute_3_min": 0.60,  
  "attribute_3_max": 0.99,  
  "attribute_4_min": 0.15,  
  "attribute_4_max": 0.98,  
  "attribute_5_min": 0.45,  
  "attribute_5_max": 0.95  
}
```

```
regional_data = [  
  {  
    "region": "United States",  
    "copper_price_per_kg": 9.50,  
    "aluminum_price_per_kg": 2.80,  
    "copper_product_market_share": 0.75,  
    "aluminum_product_market_share": 0.25,  
    "weight_attribute_1": 0.20,  
    "weight_attribute_2": 0.25,  
    "weight_attribute_3": 0.20,  
    "weight_attribute_4": 0.15,  
    "weight_attribute_5": 0.20  
  },  
  {  
    "region": "China",  
    "copper_price_per_kg": 9.80,  
    "aluminum_price_per_kg": 2.15,  
    "copper_product_market_share": 0.65,  
  }  
]
```

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    "aluminum_product_market_share": 0.35,
    "weight_attribute_1": 0.30,
    "weight_attribute_2": 0.10,
    "weight_attribute_3": 0.25,
    "weight_attribute_4": 0.20,
    "weight_attribute_5": 0.15
  },
  {
    "region": "Europe",
    "copper_price_per_kg": 9.55,
    "aluminum_price_per_kg": 2.65,
    "copper_product_market_share": 0.60,
    "aluminum_product_market_share": 0.40,
    "weight_attribute_1": 0.15,
    "weight_attribute_2": 0.20,
    "weight_attribute_3": 0.25,
    "weight_attribute_4": 0.25,
    "weight_attribute_5": 0.15
  },
  {
    "region": "India",
    "copper_price_per_kg": 10.10,
    "aluminum_price_per_kg": 2.55,
    "copper_product_market_share": 0.85,
    "aluminum_product_market_share": 0.15,
    "weight_attribute_1": 0.45,
    "weight_attribute_2": 0.15,
    "weight_attribute_3": 0.20,
    "weight_attribute_4": 0.10,
    "weight_attribute_5": 0.10
  }
]

```

```

product_data = [
  {
    "region": "United States",
    "dominant material": "cu",
    "product": "EV Wire Harness",
    "copper_kg": 22.0,
    "aluminum_kg": 2.0,
    "total_mass_kg": 45.0,
    "non_material_cost_per_unit": 350.0,
    "attribute_2_value": 0.90,
    "attribute_3_value": 0.85,

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"attribute_4_value": 0.40,
"attribute_5_value": 0.90
},
{
  "region": "United States",
  "dominant material": "al",
  "product": "EV Wire Harness",
  "copper_kg": 5.0,
  "aluminum_kg": 9.5,
  "total_mass_kg": 32.5,
  "non_material_cost_per_unit": 420.0,
  "attribute_2_value": 0.20,
  "attribute_3_value": 0.80,
  "attribute_4_value": 0.70,
  "attribute_5_value": 0.65
},
{
  "region": "China",
  "dominant material": "cu",
  "product": "EV Wire Harness",
  "copper_kg": 16.0,
  "aluminum_kg": 4.0,
  "total_mass_kg": 35.0,
  "non_material_cost_per_unit": 180.0,
  "attribute_2_value": 0.55,
  "attribute_3_value": 0.90,
  "attribute_4_value": 0.65,
  "attribute_5_value": 0.75
},
{
  "region": "China",
  "dominant material": "al",
  "product": "EV Wire Harness",
  "copper_kg": 4.0,
  "aluminum_kg": 7.0,
  "total_mass_kg": 25.5,
  "non_material_cost_per_unit": 230.0,
  "attribute_2_value": 0.65,
  "attribute_3_value": 0.85,
  "attribute_4_value": 0.85,
  "attribute_5_value": 0.60
},
{
  "region": "Europe",
```

```
"dominant material": "cu",
"product": "EV Wire Harness",
"copper_kg": 15.0,
"aluminum_kg": 5.0,
"total_mass_kg": 38.0,
"non_material_cost_per_unit": 400.0,
"attribute_2_value": 0.75,
"attribute_3_value": 0.95,
"attribute_4_value": 0.80,
"attribute_5_value": 0.85
},
{
  "region": "Europe",
  "dominant material": "al",
  "product": "EV Wire Harness",
  "copper_kg": 3.5,
  "aluminum_kg": 7.5,
  "total_mass_kg": 27.0,
  "non_material_cost_per_unit": 460.0,
  "attribute_2_value": 0.40,
  "attribute_3_value": 0.90,
  "attribute_4_value": 0.92,
  "attribute_5_value": 0.70
},
{
  "region": "India",
  "dominant material": "cu",
  "product": "EV Wire Harness",
  "copper_kg": 12.0,
  "aluminum_kg": 0.5,
  "total_mass_kg": 26.0,
  "non_material_cost_per_unit": 110.0,
  "attribute_2_value": 0.95,
  "attribute_3_value": 0.75,
  "attribute_4_value": 0.30,
  "attribute_5_value": 0.80
},
{
  "region": "India",
  "dominant material": "al",
  "product": "EV Wire Harness",
  "copper_kg": 3.0,
  "aluminum_kg": 5.5,
  "total_mass_kg": 19.0,
```

```
"non_material_cost_per_unit": 160.0,  
"attribute_2_value": 0.15,  
"attribute_3_value": 0.70,  
"attribute_4_value": 0.60,  
"attribute_5_value": 0.55  
}  
]
```